

VPA ICT — IGCSE Y10S1 Weeks 1–9: Data Analysis & Web Authoring

Weeks 1 – 9: Data Analysis & Web Authoring

Syllabus: CIE IGCSE ICT (0417) Topics 13 & 14 | Software: Excel/Calc + Text Editor + Browser

Week 1 — Advanced Spreadsheet Formulas

Syllabus: 13.1 Advanced Data Analysis

Keywords: nested IF 嵌套IF (qiàntào IF), VLOOKUP 垂直查找 (chuízhí cházhǎo), HLOOKUP 水平查找 (shuǐpíng cházhǎo), INDEX/MATCH 索引匹配 (suǒyǐn pǐpèi), array formula 数组公式 (shùzǔ gōngshì), named range 命名区域 (mìngmíng qūyù)

Activity 1 (15 min): Nested IF challenge — Create a shipping cost calculator with 4 tiers: <1kg (\$5), 1–5kg (\$10), 5–10kg (\$15), >10kg (\$20).

Activity 2 (15 min): VLOOKUP mastery — Create a product lookup: input ProductID, auto-fill Name, Price, Category from table.

Activity 3 (10 min): INDEX/MATCH — Why is it better than VLOOKUP? (Can look left! No column number issues!)

=IF(B2

Exam Tip: For lookup questions, always use FALSE/0 for exact match. TRUE/1 for approximate match (e.g., grade boundaries). The exam usually requires exact match.

Week 2 — What-If Analysis

Syllabus: 13.2 What-If Analysis

Keywords: what-if 假设分析 (jiǎshè fēnxī), scenario manager 方案管理器 (fāng'àn guǎnlǐqì), goal seek 单变量求解 (dān biànlìang qiújiě), data table 模拟运算表 (mónǐ yùnsuàn biǎo), sensitivity analysis 敏感性分析 (mǐngǎnxìng fēnxī)

Activity 1 (15 min): Goal Seek — "I want profit to be \$10,000. What should the price be?"

Activity 2 (15 min): Scenario Manager — Create 3 scenarios for a business: Best Case, Expected Case, Worst Case. Compare side-by-side.

Activity 3 (10 min): One-variable data table — Show how monthly payment changes with interest rates from 2% to 10%.

Week 3 — Advanced Charts & Pivot Tables

Syllabus: 13.3 Charts and Pivot Tables

Activity 1 (15 min): Create a combo chart with secondary axis — Sales (bars) vs. Profit Margin (line, %).

Activity 2 (15 min): Pivot table with slicers — Sales by Region and Product, filterable by Quarter.

Activity 3 (10 min): Pivot chart — Dynamic chart linked to pivot table. When pivot changes, chart updates automatically.

Week 4 — Data Import, Export & Integration

Syllabus: 13.4 Data Import/Export

Keywords: CSV 逗号分隔值 (dòuhào fēngé zhí), import 导入 (dǎorù), export 导出 (dǎochū), delimited 分隔符 (fēngé fú), text to columns 分列 (fēnliè), query 查询 (cháxún)

Activity 1 (15 min): Import CSV file into Excel. Clean data: fix dates, split full name into first/last, remove duplicates.

Activity 2 (15 min): Export Excel data as CSV and HTML. Compare file formats.

Activity 3 (10 min): Link Excel to external data — Refresh data from a CSV when it changes.

Week 5 — Web Authoring: HTML Structure

Syllabus: 14.1 Web Page Basics

Keywords: HTML 超文本标记语言 (chāo wénběn biāojiǐ yǔyán), tag 标签 (biāoqiān), element 元素 (yuánsù), attribute 属性 (shǔxìng), DOCTYPE 文档类型 (wéndǎng lèixíng), head 头部 (tóubù), body 主体 (zhǔtǐ), title 标题 (biāotí)

Activity 1 (15 min): "Perfect HTML Skeleton" — Write a complete, valid HTML5 document from memory:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>My Page</title>

</head>

<body>

<h1>Welcome</h1>

<p>This is my first paragraph.</p>

</body>

</html>
```

Activity 2 (15 min): Validate HTML using validator.w3.org. Fix any errors.

Activity 3 (10 min): Semantic HTML — Use header, nav, main, article, footer instead of generic divs.

Exam Tip: HTML questions in Paper 3 often show code with errors. Check for: missing closing tags, wrong nesting, missing DOCTYPE, missing alt attributes on images, missing charset.

Week 6 — CSS Styling

Syllabus: 14.2 Cascading Style Sheets

Keywords: CSS 层叠样式表 (céngdié yàngshì biǎo), selector 选择器 (xuǎnzéqì), class 类 (lèi), id 标识符 (biāoshí fú), property 属性 (shǔxìng), value 值 (zhí), external CSS 外部样式表 (wàibù yàngshì biǎo), box model 盒模型 (hé móxíng)

Activity 1 (15 min): Create external stylesheet (style.css). Link to HTML. Style h1, p, body.

Activity 2 (15 min): Box model practice — Create a div with: width 300px, padding 20px, border 5px, margin 10px. Calculate total width.

Activity 3 (10 min): Class vs. ID — When to use each? (class for repeating styles, id for unique elements)

```
/* External CSS */

body { font-family: Arial; background-color: #f0f0f0; }

.highlight { color: red; font-weight: bold; }
```

```
#header { width: 100%; height: 100px; background: navy; }
```

Week 7 — Web Page Layout: Tables & Lists

Syllabus: 14.3 Tables, Lists, and Images

Keywords: table 表格 (biǎogé), tr 表格行 (biǎogé háng), td 表格单元格 (biǎogé dānyuán gé), th 表头单元格 (biǎotóu dānyuán gé), colspan 跨列 (kuà liè), rowspan 跨行 (kuà háng), ul/ol/li 无序/有序列表 (wúxù/yǒuxù lièbiǎo)

Activity 1 (15 min): Create a timetable using HTML table with colspan and rowspan for merged cells.

Activity 2 (15 min): Nested lists — Create a multi-level navigation menu using ul and li.

Activity 3 (10 min): Image optimization — Save same image as JPG (quality 80, 60, 40) and PNG. Compare file sizes and visual quality.

Week 8 — Hyperlinks, Navigation & Forms

Syllabus: 14.4 Hyperlinks and Navigation

Keywords: hyperlink 超链接 (chāo liànjiē), anchor 锚点 (máodiǎn), href 链接地址 (liànjiē dìzhǐ), target 目标 (mùbiāo), form 表单 (biǎodān), input 输入框 (shūrù kuàng), textarea 文本域 (wénběn yù), button 按钮 (ànniǔ)

Activity 1 (15 min): Create navigation bar — 5 links to other pages in the site. Test all links work.

Activity 2 (15 min): Contact form — Create form with: name (text), email (email), message (textarea), submit (button).

Activity 3 (10 min): Internal links — Create "Back to Top" links and anchor links to sections on the same page.

```
<form action="submit.php" method="post">
```

```
Name: <input type="text" name="name" required><br>
```

```
Email: <input type="email" name="email"><br>
```

```
<textarea name="message" rows="4" cols="50"></textarea><br>
```

```
<input type="submit" value="Send">
```

```
</form>
```

Exam Tip: For form questions, check: 1) correct input types (text, email, password, radio, checkbox), 2) name attributes present, 3) method (GET vs POST), 4) submit button included. Missing any loses marks.

Week 9 — Web Design Project & Assessment

Syllabus: 14.5 Web Design Project

Project (40 min): Create a 4-page website for a school club or local business:

index.html — Homepage with navigation, welcome message, image

about.html — About page with styled text and list

products.html — Products/services with HTML table and images

contact.html — Contact form with at least 4 different input types

Requirements:

External CSS file linked to all pages

Consistent navigation on every page

At least one table with merged cells

Form with text, email, textarea, and submit

All images have alt attributes

Valid HTML5 (test with validator)

Marking:

Bronze: 3 pages, basic HTML, some CSS, working links

Silver: 4 pages, external CSS, table, form, consistent design

Gold: All above + semantic HTML, optimized images, responsive elements, validated code

Integration Focus: Data → Web

Challenge: Export spreadsheet data as HTML table. Insert that table into your website. Update spreadsheet → re-export → update website. This mirrors Paper 3 integration tasks.